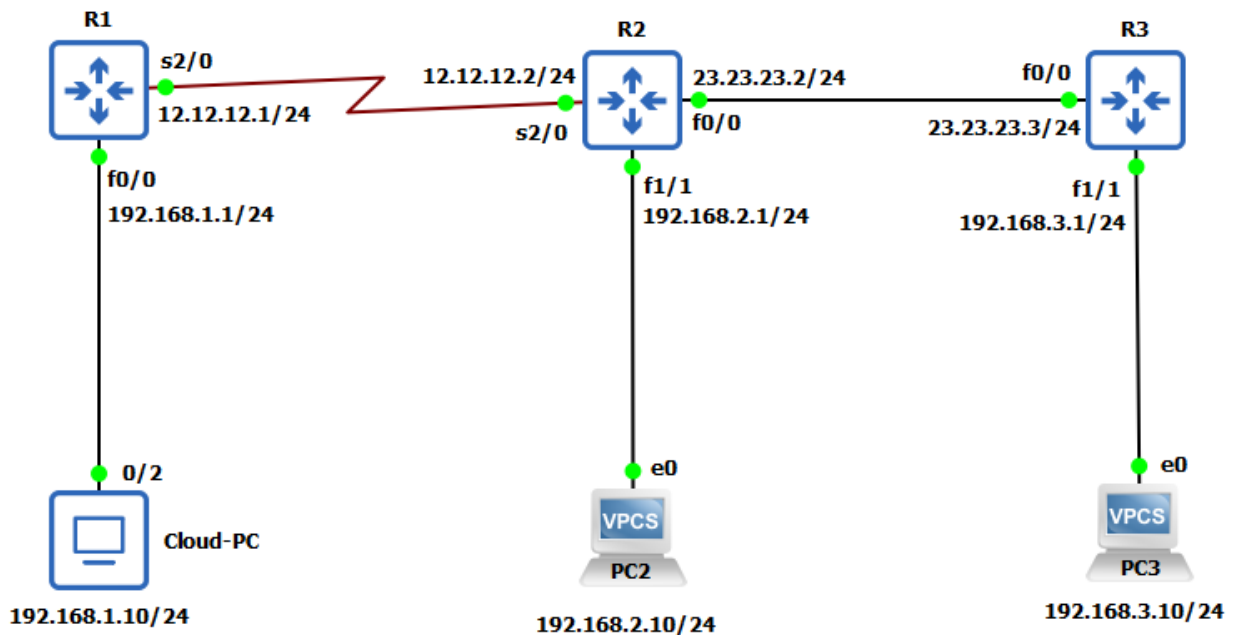


Static Routing



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** and device access password as **cisco** (Refer Lab-1)
2. Configure static routes (next-hop / exit interface) on routers to provide network reachability.
3. Cloud-PC, PC-2 & PC-3 should ping each other.

Note: Cloud-PC is a cloud which bridge the R1 router with loopback adapter on real PC.

Static routing on R1 Router

```
R1#config t
R1(config)#ip route 23.23.23.0 255.255.255.0 12.12.12.2
R1(config)#ip route 192.168.2.0 255.255.255.0 12.12.12.2
R1(config)#ip route 192.168.3.0 255.255.255.0 12.12.12.2
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#ip route 192.168.1.0 255.255.255.0 12.12.12.1
R2(config)#ip route 192.168.3.0 255.255.255.0 23.23.23.3
R2(config)#exit
R2#
```

Static routing on R3 Router

```
R3#config t
R3(config)#ip route 12.12.12.0 255.255.255.0 23.23.23.2
R3(config)#ip route 192.168.1.0 255.255.255.0 23.23.23.2
R3(config)#ip route 192.168.2.0 255.255.255.0 23.23.23.2
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh ip route**

Codes: L - local, C - connected, **S - static**, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.12.12.0/24 is directly connected, Serial2/0
L    12.12.12.1/32 is directly connected, Serial2/0
23.0.0.0/24 is subnetted, 1 subnets
S   23.23.23.0 [1/0] via 12.12.12.2
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
S   192.168.2.0/24 [1/0] via 12.12.12.2
S   192.168.3.0/24 [1/0] via 12.12.12.2
R1#
```

R2#sh ip route static

Codes: L - local, C - connected, **S - static**, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

```
S   192.168.1.0/24 [1/0] via 12.12.12.1
S   192.168.3.0/24 [1/0] via 23.23.23.3
R2#
```

R3#sh ip route static | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

S 12.12.12.0 [1/0] via 23.23.23.2

S 192.168.1.0/24 [1/0] via 23.23.23.2

S 192.168.2.0/24 [1/0] via 23.23.23.2

R3#

R1#sh protocols | exclude down

Global values:

Internet Protocol routing is enabled

FastEthernet0/0 is up, line protocol is up

Internet address is 192.168.1.1/24

Serial2/0 is up, line protocol is up

Internet address is 12.12.12.1/24

R2#sh protocols | exclude down

Global values:

Internet Protocol routing is enabled

FastEthernet0/0 is up, line protocol is up

Internet address is 23.23.23.2/24

FastEthernet1/1 is up, line protocol is up

Internet address is 192.168.2.1/24

Serial2/0 is up, line protocol is up

Internet address is 12.12.12.2/24

R3#sh protocols | exclude down

Global values:

Internet Protocol routing is enabled

FastEthernet0/0 is up, line protocol is up

Internet address is 23.23.23.3/24

FastEthernet1/1 is up, line protocol is up

Internet address is 192.168.3.1/24

Ping PC2 & PC3 from R1**R1#ping 192.168.2.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/84 ms

R1#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 76/84/112 ms

R1#

Ping Cloud-PC & PC3 from R2**R2#ping 192.168.1.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/58/80 ms

R2#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/55/60 ms

R2#

Ping Cloud-PC & PC2 from R3**R3#ping 192.168.1.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 68/84/104 ms

R3#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 76/78/80 ms

R3#

Ping PC2 & PC3 from Cloud-PC**C:\Users\Ratan>ping 192.168.2.10**

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=3039ms TTL=62

Reply from 192.168.2.10: bytes=32 time=34ms TTL=62

Reply from 192.168.2.10: bytes=32 time=36ms TTL=62

Reply from 192.168.2.10: bytes=32 time=39ms TTL=62

Ping statistics for 192.168.2.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 34ms, Maximum = 3039ms, Average = 787ms

C:\Users\Ratan>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=3050ms TTL=61

Reply from 192.168.3.10: bytes=32 time=56ms TTL=61

Reply from 192.168.3.10: bytes=32 time=60ms TTL=61

Reply from 192.168.3.10: bytes=32 time=60ms TTL=61

Ping statistics for 192.168.3.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 56ms, Maximum = 3050ms, Average = 806ms

Ping Cloud-PC & PC3 from PC2

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=126 time=43.200 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=126 time=42.535 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=126 time=34.526 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=126 time=36.341 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=126 time=42.530 ms

PC2> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=62 time=36.183 ms

84 bytes from 192.168.3.10 icmp_seq=2 ttl=62 time=43.534 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=36.352 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=39.167 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=41.213 ms

Ping Cloud-PC & PC2 from PC3

PC3> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=125 time=64.098 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=125 time=63.022 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=125 time=55.364 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=125 time=56.361 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=125 time=65.387 ms

PC3> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=39.457 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=38.493 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=37.288 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=37.205 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=35.614 ms

Trace the path for PC2 & PC3 from Cloud-PC

C:\Users\Ratan>tracert 192.168.3.10

Tracing route to DESKTOP-CD66NV2 [192.168.3.10]
over a maximum of 30 hops:

1	7 ms	9 ms	10 ms	192.168.1.1
2	32 ms	29 ms	30 ms	12.12.12.2
3	52 ms	52 ms	53 ms	23.23.23.3
4	3057 ms	62 ms	61 ms	DESKTOP-CD66NV2 [192.168.3.10]

Trace complete.

C:\Users\Ratan>tracert 192.168.2.10

Tracing route to DESKTOP-CD66NV2 [192.168.2.10]
over a maximum of 30 hops:

```

 1  10 ms  10 ms  8 ms  192.168.1.1
 2  31 ms  31 ms  30 ms  12.12.12.2
 3  34 ms  40 ms  39 ms  DESKTOP-CD66NV2 [192.168.2.10]

```

Trace complete.

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.1	-	ca01.2088.0008	ARPA	FastEthernet0/0
Internet	192.168.1.10	0	0200.4c4f.4f50	ARPA	FastEthernet0/0

R1#

R2#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	23.23.23.2	-	ca02.0c24.0008	ARPA	FastEthernet0/0
Internet	23.23.23.3	7	ca03.1e00.0008	ARPA	FastEthernet0/0
Internet	192.168.2.1	-	ca02.0c24.001d	ARPA	FastEthernet1/1
Internet	192.168.2.10	7	0050.7966.6801	ARPA	FastEthernet1/1

R2#

R3#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	23.23.23.2	7	ca02.0c24.0008	ARPA	FastEthernet0/0
Internet	23.23.23.3	-	ca03.1e00.0008	ARPA	FastEthernet0/0
Internet	192.168.3.1	-	ca03.1e00.001d	ARPA	FastEthernet1/1
Internet	192.168.3.10	7	0050.7966.6800	ARPA	FastEthernet1/1

R3#

R1#sh ip route 192.168.3.10

Routing entry for 192.168.3.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 12.12.12.2

Route metric is 0, traffic share count is 1

R1#sh ip route 192.168.2.10

Routing entry for 192.168.2.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 12.12.12.2

Route metric is 0, traffic share count is 1

R2#sh ip route 192.168.1.10

Routing entry for 192.168.1.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 12.12.12.1

Route metric is 0, traffic share count is 1

R2#sh ip route 192.168.3.10

Routing entry for 192.168.3.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 23.23.23.3

Route metric is 0, traffic share count is 1

R3#sh ip route 192.168.1.10

Routing entry for 192.168.1.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 23.23.23.2

Route metric is 0, traffic share count is 1

R3#sh ip route 192.168.2.10

Routing entry for 192.168.2.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 23.23.23.2

Route metric is 0, traffic share count is 1

PC2> sh arp

ca:02:0c:24:00:1d 192.168.2.1 expires in 56 seconds

PC3> sh arp

ca:03:1e:00:00:1d 192.168.3.1 expires in 112 seconds

C:\Users\Ratan>arp -a

Interface: 192.168.1.10 --- 0x18

Internet Address	Physical Address	Type
192.168.1.1	ca-01-20-88-00-08	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static

Remove the previous configured static route and reconfigure the static route with self-exit interface.

Static routing on R1 Router

```
R1#config t
R1(config)#no ip route 23.23.23.0 255.255.255.0 12.12.12.2
R1(config)#no ip route 192.168.2.0 255.255.255.0 12.12.12.2
R1(config)#no ip route 192.168.3.0 255.255.255.0 12.12.12.2
R1(config)#ip route 23.23.23.0 255.255.255.0 s2/0
R1(config)#ip route 192.168.2.0 255.255.255.0 s2/0
R1(config)#ip route 192.168.3.0 255.255.255.0 s2/0
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#no ip route 192.168.1.0 255.255.255.0 12.12.12.1
R2(config)#no ip route 192.168.3.0 255.255.255.0 23.23.23.3
R2(config)#ip route 192.168.1.0 255.255.255.0 s2/0
R2(config)#ip route 192.168.3.0 255.255.255.0 fa0/0
R2(config)#exit
R2#
```

Static routing on R3 Router

```
R3#config t
R3(config)#no ip route 12.12.12.0 255.255.255.0 23.23.23.2
R3(config)#no ip route 192.168.1.0 255.255.255.0 23.23.23.2
R3(config)#no ip route 192.168.2.0 255.255.255.0 23.23.23.2
R3(config)#ip route 12.12.12.0 255.255.255.0 fa0/0
R3(config)#ip route 192.168.1.0 255.255.255.0 fa0/0
R3(config)#ip route 192.168.2.0 255.255.255.0 fa0/0
R3(config)#exit
R3#
```

✓ Verification & Testing:

R1#sh ip route static | begin Gateway

Gateway of last resort is not set

23.0.0.0/24 is subnetted, 1 subnets

```
S    23.23.23.0 is directly connected, Serial2/0
S    192.168.2.0/24 is directly connected, Serial2/0
S    192.168.3.0/24 is directly connected, Serial2/0
R1#
```


R2#sh ip route static | begin Gateway

Gateway of last resort is not set

S 192.168.1.0/24 is directly connected, Serial2/0
S 192.168.3.0/24 is directly connected, FastEthernet0/0
R2#

R3#sh ip route static | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets
S 12.12.12.0 is directly connected, FastEthernet0/0
S 192.168.1.0/24 is directly connected, FastEthernet0/0
S 192.168.2.0/24 is directly connected, FastEthernet0/0
R3#

Ping PC2 & PC3 from R1**R1#ping 192.168.2.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/84 ms

R1#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 76/84/112 ms

R1#

Ping Cloud-PC & PC3 from R2**R2#ping 192.168.1.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/58/80 ms

R2#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/55/60 ms

R2#

Ping Cloud-PC & PC2 from R3

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 68/84/104 ms

R3#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 76/78/80 ms

R3#

Ratan

Important Commands:

```
sh ip route  
sh ip route static  
sh ip route static | begin Gateway  
sh protocols | exclude down  
sh arp
```

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